PCBC H Parametric & Sequences Test (Practice C) **Name**_____ _____/27

SHOW ALL WORK FOR FULL CREDIT. ***Calculator*** **You have the entire period.**

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this test.

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_____ _____

Signature

Solve the following application problem. Report answers to the thousandths. Include units in your answers. (4 points)

7. An archer shoots an arrow from a height of 6 feet at an angle of 25° with an initial speed of 150 ft/s.

a) Write a set of parametric equations for the motion of the arrow. (Recall:
$x = (v\cos\theta)t,\ \ y = (v\sin\theta)t - 16t^2 + h$)

b) Determine the maximum height reached by the arrow.

c) Determine how long the arrow is in the air.

d) Determine how far the arrow traveled in the air.

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern. (3 points)

8. $-\frac{1}{2}, \frac{4}{3}, -\frac{9}{4}, \frac{16}{5}, \ldots$

Write out the sum. (3 points)

9. $\sum_{k=1}^{n} \frac{2k-1}{k}$

Find each sum. Show the properties and formulas you use, then use your calculator. (#10 and #11 are 3 points each, #12 is 4 points)

10. $\sum_{k=1}^{25} (4k-3)$

11. $\sum_{k=1}^{12} k(k+1)$

12. $\sum_{k=7}^{18} \left(k^2+5\right)$

Expand using the Binomial Theorem. (#13 is 3 points, #14 is 4 points)

13. $(x+1)^8$

14. $(x-3)^5$

Extra practice: more sequence and summation questions

Not part of the 27 points. Worked in the key.

Express the sum using summation notation.

E1. $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \cdots + 15 \cdot 16$

The given pattern continues. Write down the nth term of the sequence $\{a_n\}$ suggested by the pattern.

E2. $3,\ -6,\ 12,\ -24,\ \ldots$

Answer Key

Work on the left, the reason beside it. Boxed = final answer.

Day 2: Calculator

7. Arrow: $h = 6$ ft, $\theta = 25°$, $v = 150$ ft/s. Calculator in **degree** mode, Graph in **Parametric**.

a) $x = (150\cos 25°)t$ — *plug v and θ into the given formula; degree symbol required*

$y = (150\sin 25°)t - 16t^2 + 6$ — *+h: starts at 6 ft*

$x_1(t) = 150\cos(25°)\cdot t,\ \ y_1(t) = 150\sin(25°)\cdot t - 16t^2 + 6$ — *what you type in*

$0 \le t \le 5$, tstep 0.01; x : 0 to 600, y : 0 to 80 — *graph doesn't come back down? raise tmax*

b) $\boxed{y \approx 68.791 \text{ ft}}$ — *trace to the top ($t \approx 1.981$); not Analyze, Maximum*

c) $\boxed{t \approx 4.055 \text{ s}}$ — *trace to $y \approx 0$; trace step 0.01, then 0.001*

d) $\boxed{x \approx 551.199 \text{ ft}}$ — *same point, read x*

Units: feet (the speed was ft/s).

8. $-\frac{1}{2}, \frac{2}{3}, -\frac{3}{4}, \frac{4}{5}, \ldots$

top: n^2, bottom: $n+1$ — *write 1 2 3 4 above; top and bottom separately*

$-,+,-,+ \Rightarrow (-1)^n$ — *first term negative*

$\boxed{a_n = (-1)^n \frac{n^2}{n+1}}$ — *check $n = 1$: $-\frac{1}{2}$*

9. $\sum_{k=1}^{n} \frac{2k-1}{k}$

$\frac{2(1)-1}{1} + \frac{2(2)-1}{2} + \frac{2(3)-1}{3} + \cdots + \frac{2n-1}{n}$ — *$k = 1,2,3$, then $k = n$*

$\boxed{1 + \frac{3}{2} + \frac{5}{3} + \cdots + \frac{2n-1}{n}}$ — *don't add*

10.

$4\sum_{k=1}^{25} k - \sum_{k=1}^{25} 3$ — *(3), (1)*

$4\left[\frac{25(26)}{2}\right] - 3(25)$ — *(6), (5)*

$\boxed{1225}$

11.

$\sum_{k=1}^{12} \left(k^2 + k\right)$ — *multiply out first*

$\sum_{k=1}^{12} k^2 + \sum_{k=1}^{12} k$ — *(2)*

$\frac{12(13)(25)}{6} + \frac{12(13)}{2}$ — *(7), (6)*

$\boxed{728}$

12.

$\sum_{k=1}^{18} (k^2+5) - \sum_{k=1}^{6} (k^2+5)$ — *(4): starts at 7, so $j = 6$*

$\left[\frac{18(19)(37)}{6} + 5(18)\right] - \left[\frac{6(7)(13)}{6} + 5(6)\right]$ — *(2), (7), (5) on each piece*

$\boxed{2078}$

13. $(x+1)^8$: $n=8$, row 1 8 28 56 70 56 28 8 1. The sign goes with its term.

$1x^8(1)^0 + 8x^7(1)^1 + 28x^6(1)^2 + 56x^5(1)^3$ — *blanks first: first term's powers count down, second term's count up*

$+70x^4(1)^4 + 56x^3(1)^5 + 28x^2(1)^6 + 8x^1(1)^7$

$+1x^0(1)^8$

$\boxed{x^8 + 8x^7 + 28x^6 + 56x^5 + 70x^4 + 56x^3 + 28x^2 + 8x + 1}$ — *second term positive: every term positive*

14. $(x-3)^5$: $n=5$, row 1 5 10 10 5 1. The sign goes with its term.

$1x^5(-3)^0 + 5x^4(-3)^1 + 10x^3(-3)^2 + 10x^2(-3)^3$ — *blanks first: first term's powers count down, second term's count up*

$+5x^1(-3)^4 + 1x^0(-3)^5$

$\boxed{x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243}$ — *second term negative: odd powers of -3 are negative, so signs alternate*

Extra practice (not scored)

E1. $1\cdot 2 + 2\cdot 3 + \cdots + 15\cdot 16$

$k(k+1)$, $k=1$ to 15 — *first factor counts 1 to 15*

$\boxed{\sum_{k=1}^{15} k(k+1)}$

E2. $\overset{1}{3}, \overset{2}{-6}, \overset{3}{12}, \overset{4}{-24}, \ldots$

top: $3\cdot 2^{n-1}$, bottom: 1 — *write 1 2 3 4 above; top and bottom separately*

$+,-,+,- \Rightarrow (-1)^{n+1}$ — *first term positive*

$\boxed{a_n = 3(-2)^{n-1}}$ — *each term is -2 times the last; $(-1)^{n+1}3\cdot 2^{n-1}$ is the same*